

- Thus, the relative importance for yield spread volatility of factors other than the overall yield level is an increasing function of the distance between the CTD yield and the notional coupon of the bond futures contract. Consequently, there is a tendency for one-factor DO models to understate the value of the DO when the yield of the CTD is far from the notional coupon.

As a result, when yields dropped globally toward levels far below the notional coupons, the one-factor DO models working well at a 6% yield level became less useful. Since most market participants did not adjust their models to reflect the new source of DO value, the net bases quoted in the market nowadays systematically underestimate the real value of the DO (i.e. tend to be too low).

As an illustration, we consider the source of yield spread volatility between deliverables in a low-yield environment for the case of JGB futures, for which this situation first was relevant. While the deliverable basket extends from seven to 10.5 years, the low yield level in Japan means that the bond with the shortest maturity is always clearly CTD and that there is practically no chance for a bond with a longer maturity to become CTD, as seen in Figure 7.1. However, there are often two or even three bonds with the same shortest maturity and only a minimal difference in coupon, thus with almost the same BPV. These two or three bonds can all easily become CTD, thus there can be CTD switches between the two or three candidates with the same maturity. In fact, because the CTD candidates are so similar, CTD switches between them are quite common. Seen through the perspective of a one-factor model, a change in the overall yield level impacts all CTD candidates in almost the same manner, given the almost identical BPV and sensitivity. Thus, even if a CTD switch occurs, the profit is negligible, as argued above. In other words, while the similarity of CTD candidates may make CTD switches between them likely, the similarity of converted BPVs also means that the impact of a change in the overall yield level on the yield spread volatility between them is only marginal. Hence, applying a one-factor model to the basket of JGB futures returns a DO value close to zero.

In reality, however, the yield spread volatility between CTD candidates can be quite large, in part because the delivery situation can magnify the natural yield spread volatility. Imagine that many investors have a short CTD net basis position on their books. If the CTD switches to another candidate, which initially may be just a little bit cheaper, they need to be concerned about being delivered the new CTD into their long futures position. Thus, in order to hedge their risk, they need to sell the new and buy back the old CTD, an operation that may magnify an initially small yield spread between the two CTD candidates and may cause more investors to close their positions, leading to a

reinforcing cycle. Note that this potentially significant yield spread volatility between deliverables is not caused and not even accompanied by any change in the overall yield level. Hence, by construction it cannot contribute toward the DO value in the context of a one-factor model, even though a long net basis position could yield significant profit in this scenario.⁵

When yields are far below the notional coupon, the yield spread volatility between deliverables is therefore not a function of the 95% of overall yield volatility explained by the overall yield level (and captured by one-factor models) but rather of the 5% of overall yield volatility not explained by the overall yield level (and not captured by one-factor models). Hence, at current yield levels, it is the 5% of overall yield volatility explained by greater factors (2, 3, ...), which matters for the yield spread volatility between deliverables. Consequently, a DO model claiming applicability in any yield environment needs to take more factors into account. DO modeling therefore needs to respond to the drop in yield levels far below the notional coupon by moving from one-factor to multi-factor models.

A FLEXIBLE MULTI-FACTOR DELIVERY OPTION MODEL⁶

Since it is impossible to determine *ex ante* which factor will be responsible for yield spread volatility between deliverables, we advise using *all* factors. That is, we replace the one-factor model by a model that has as many factors as there are bonds in the deliverable basket. The full information needed to assess the yield spread volatility between deliverables and thus the value of the DO can be represented for a deliverable basket with n bonds with the yields $y_1, \dots, y_n$ by the variables

- y_1 : the absolute yield level of the first deliverable bond. The designation 'first' is arbitrary. For example, the bond with the shortest maturity in the basket could be considered the first bond, as could the CTD.
- $y_2 - y_1, \dots, y_n - y_1$ (i.e. the yield spread between every other bond and the first one).

⁵This is also the reason why an arbitrage between the net basis and futures options is only practicable when CTD switches between the short and long end of the basket are the main driving force of the net basis.

⁶That model was developed by us as employees of ABN Amro and first published in the ABN Amro research note "Exploiting the ignored delivery option in JGB contracts" from 21 February 2002. It is reproduced here with kind permission from RBS.

Note that this set of variables is just a different way of modeling all yields $y_1, \dots, y_n$. The reason for this particular representation will become clear later on. While we can make reasonable assumptions about how yield spread variables behave in a delivery situation, expressing that impact of a futures contract on the relationships between deliverable bonds in the form of correlations between $y_1, \dots, y_n$ would be a major and unnecessary challenge.

We allow for correlation between all of the variables (i.e. both between the absolute yield level and yield spread variables⁷ and between different yield spread variables). Furthermore, we assume all variables are normally distributed. Note that this is the only assumption of the model and in our view not a very strong or particularly restrictive assumption. If a particular market was found to behave differently, the following simulation could be adjusted accordingly. (For example, one could replace the normal distribution with a lognormal distribution.) We can now obtain an estimate for the DO through the steps shown in Box 7.2.

Box 7.2

Multi-Factor Delivery Option Model

- Define the yield volatility of the first deliverable bond.
- Define the yield spread volatility for the variables $y_2 - y_1, \dots, y_n - y_1$.
- Define the correlations between all of the variables $y_1, y_2 - y_1, \dots, y_n - y_1$. We shall discuss ways to set these input parameters below.
- Run a Monte Carlo simulation with normally distributed random variables each with zero mean and with standard deviations and correlations defined as above, simulating the evolution of the yield (spreads) of all deliverable bonds until the delivery date.
- Translate the results of each simulation into a table of yields $y_1, \dots, y_n$ at delivery and calculate the corresponding converted prices. In order to adjust for the difference between forward and spot prices, add the difference between the current converted forward price and the average of the future converted prices at delivery generated by the simulation to each of the simulation results. In this way, the mean of the simulated prices on the delivery date for each bond will equal the forward price for each bond.

⁷The yield beta of a one-factor model would show up via the covariance matrix in the multifactor model.

- For each of the mean-adjusted simulation results, identify the bond that is CTD and the extent of any profit from a CTD switch.
- Calculate an estimate for the DO value as the average of the profit from CTD switches over all simulations, taking the CFs into account.
- Calculate for each deliverable bond an estimate for the probability of being CTD at delivery date by dividing the number of simulations in which it finishes as CTD by the number of all simulations.

Based on these results, one can repeat the calculation of the fair value of the future and related numbers from above with a better estimation for the value of the DO. In particular:

- Calculate the fair futures price as $\frac{FwdP_{CTD}}{CF_{CTD}} - DO$.
- Repeating the exercise for the back month contract, calculate the fair futures roll (calendar spread).
- Calculate the OABPV (with reference to the first bond) by repeating the Monte Carlo simulation on a set of yields y_i as starting points, which reflects the typical impact of a 1 bp increase in y_1 on the yields of all deliverables. Then, the OABPV is the difference between the fair futures price and the fair futures price of the simulation starting with the shifted yields. As required, this calculation takes the impact of a different yield level on the DO into account.

One way to obtain the set of yields y_i , which reflects the typical impact of a 1 bp increase in y_1 on the yields of all deliverables, is to perform a PCA on the covariance matrix given by the input parameters. Then, the sensitivities to the first factor show how a 1 bp increase of y_1 is expected to influence the yields of the other deliverables. This is an application of the curve reconstruction technique through PCA outlined in Chapter 3.

A good graphical representation of both the input parameters into that Monte Carlo simulation and its output can be obtained by using the change in the yield spread variables until delivery as coordinates and plotting the resulting CTD of each simulation as a specific mark (such as points for the first bond, crosses for the second, etc.) into that coordinate system. An example for such a chart is given in Figure 7.3, which provides immediate intuition of the scenarios under which each bond is likely to be CTD.

The shape of the scatterplot in Figure 7.3 is determined by the two yield spread volatilities and by the correlation between the two yield spreads.

The CTD probability for each bond corresponds to the relative frequency of the symbols in the scatterplot depicted in Figure 7.3. Also, by assessing the

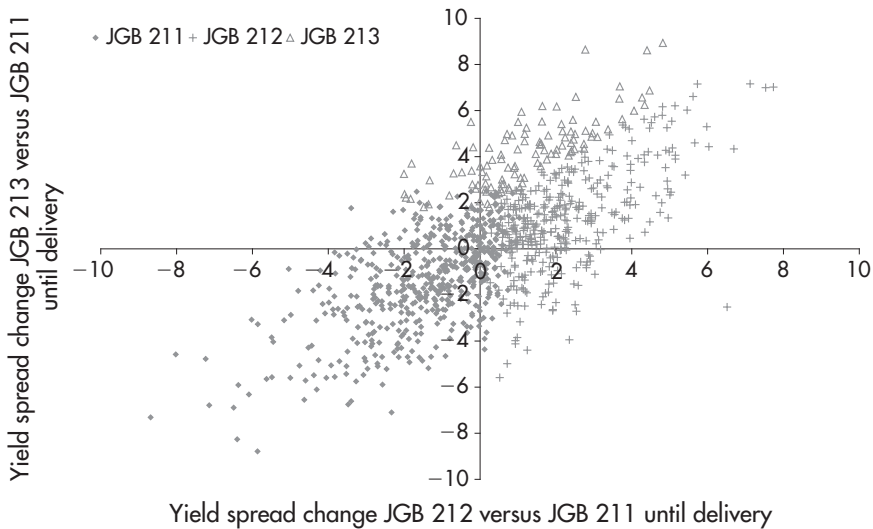


FIGURE 7.3 Results of a Monte Carlo simulation of the CTD situation at delivery date as a function of the yield spread changes between CTD candidates until delivery. *Source:* ABN Amro (reproduced here with permission from RBS).

positions of the symbols, one can get an understanding of the way a shift in yield spreads influences the CTD situation or, conversely, which yield spread move would need to happen in order for a particular bond to become CTD.

Choosing Input Parameters for the Multi-Factor DO Model

After outlining the model, we now discuss ways to choose its input parameters. The yield volatility, which is the only parameter involved in one-factor models, can be set equal to the implied volatility of options on bond futures. With regards to defining the parameters for yield spread volatilities, we face a basic problem in that historical yield spread volatility is of limited value.

Outside of the context of a deliverable basket, there is usually little reason for very similar bonds to exhibit significant yield spread volatilities. For example, why should an investor's preference for JGB 212 1.5% maturing on 22 June 2009 and JGB 213 1.4%, also maturing on 22 June 2009, suddenly change? In fact, these very similar issues are not considered as *individual* bonds but rather as almost identical representations of the same issue. Therefore, before becoming relevant for delivery into a futures contract, the yield spread volatility between deliverables is typically small.

But this situation changes when the bonds enter into competition for CTD status. JGB 212 might be CTD, but JGB 213 is not, which has a major impact on the treatment of both issues in the repo market and in basis trades. We have given an example earlier of the way a CTD switch can impact the yield spread between the two deliverable bonds involved. Hence, bonds gain an individuality, an individual treatment in the bond and repo markets, through the context of a deliverable basket. And consequently, the CTD situation is a major (most of the time the only) reason for a significant yield spread volatility between otherwise very similar bonds. Figure 7.4 shows the way the yield spread volatilities between CTD candidates increase as the delivery date approaches. On average, the yield spread volatility more than doubles in the year prior to delivery, reflecting the strong impact of a delivery situation on *individual* bonds versus their peers. In the case of a squeeze situation, the increase of yield spread volatility can be far above that in typical circumstances.

The calculation of the DO value through a multi-factor model requires an input parameter for the yield spread volatilities before delivery (i.e. before the actual impact of the specific delivery situation on the yield spread volatilities can be known). In addition, that input parameter has a significant importance for the resulting DO value, which (in case of a large distance between the

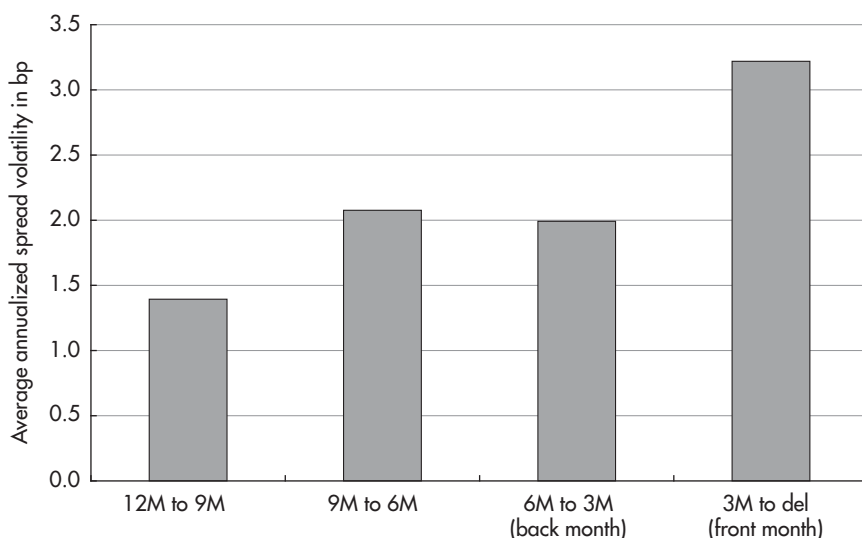


FIGURE 7.4 Evolution of the yield spread volatility between CTD candidates as they approach delivery in the example of JGB futures contracts.

Source: ABN Amro (reproduced here with permission from RBS).

actual yield level and the notional coupon) often depends almost exclusively on that input variable. Actually, drawing the estimated DO value as a function of the yield spread volatility used in the Monte Carlo simulation often reveals an almost linear relationship, with the line crossing through the origin of the coordinate system.

A possible solution to this problem could be first to calculate the historical yield spread volatility between the deliverable issues before they became part of the deliverable basket and then to adjust that yield spread volatility by the factor of its average increase into delivery, as shown in Figure 7.4. This main or average scenario could further be complemented by alternative scenarios, for example, reflecting the expected increase of yield spread volatility in case of a CTD squeeze. As a general remark, it is advisable to run the DO model described above several times for a set of different input parameters in order to gain an insight into the value of the DO and thus the expected performance of related trades (such as net basis positions) under various scenarios.

As an alternative⁸ to using constant values for the yield spread volatilities throughout the simulation, for example, by adjusting the historical yield spread volatilities by a factor derived from Figure 7.4, one can also reflect the expected increase in yield spread volatility as delivery increases directly in the simulation. In this case, the evolution of yield spread volatility shown in Figure 7.4 is applied to every path of the simulation behind Figure 7.3.

All the correlations in this multi-factor model usually remain stable in the context of a delivery, unlike the yield spread volatilities themselves. Moreover, in many cases, the resulting DO value is not significantly affected by a small change in the correlation parameters. For the purpose of the DO model, it is therefore usually fine to use the historically observed correlations as input variables. Still, as the impact of changes to the correlation parameters could be exceptionally strong in a particular delivery situation, we recommend checking the results of the DO model also for alternative correlation scenarios, just as described above for alternative yield spread volatility parameters.

The reward for these efforts is the ability to detect DOs that are undervalued in a market whose participants stick mostly to one-factor models. In the example of the JGB futures contract of Figure 7.3, the one factor model outlined at the beginning of this chapter returned a DO of less than 1 sen, while the multi-factor DO model⁹ suggests it is in fact worth 7 sen. Thus, the methodological progress can be exploited by finding and buying valuable DOs at a

⁸Rubin Rajendram has pointed out this alternative to us.

⁹Using the same yield volatility as the one-factor model and historical yield spread volatilities and correlations, with the yield spread volatilities adjusted by the factor from Figure 7.4.

bargain price. As long as the majority of market participants continue using one-factor models, the profit from buying undervalued DOs (e.g. through long CTD net basis positions) must come from actual CTD switches. The downside in case there is no CTD switch is very limited if the DO can be bought almost for free, as in the JGB futures example above. In addition, however, the increasing application of multifactor models should result in a general trend, correcting the current undervaluation of DOs toward an overall increase in net bases.

In summary, a better estimation for the DO value allows a better estimation for the fair value of the futures contract, which enhances:

- basis trading
- hedging of bonds with the future (using the OABPV)
- rolling over from the front in the back month contract.

Fitted Bond Curves

INTRODUCTION

Which of an issuer's bonds are rich, and which are cheap? And how do we know? To answer these questions, we first need to better define the terms rich and cheap in this context.

Not surprisingly given that our subject matter is relative value, we'll define these words in relative terms rather than in absolute terms. In particular, we say that a bond is rich if its price is greater than we'd expect conditional on the available information, including but not limited to the prices of the issuer's other bonds. Other information might include whether the bond is special in the repo market, whether the bond is deliverable into a futures contract, and the liquidity of the bond as proxied by the bid-ask spread.

FRAMEWORK OF ANALYSIS

Whenever we're dealing with expectations conditional on a set of information, we're in the realm of regression analysis. In particular, at any point in time, we'll specify the price of each bond in our data set as the dependent variable and the relevant conditioning information as the independent variables, as in the cross-sectional regression given by:

$$P_i = A(\Theta_i; \alpha) + B(\Psi_i; \gamma) + \varepsilon_i$$

where P_i is the price of the i -th bond in our data set, $i = 1, \dots, N$; $A(\Theta_i; \alpha)$ is a function of the parameter vector α , and of Θ_i , an array of dimension $2 \times M_i$ that represents the dates and sizes of the M_i cash flows that constitute bond i ; $B(\Psi_i; \gamma)$ is a function of the $k \times 1$ parameter vector, γ , and of the $k \times 1$ vector Ψ_i , which contains the independent variables other than the cash flow dates and amounts on which we condition our expectations of the bond prices; and ε_i is a random error term, unique to each bond in our data set.